

Frozen gdTAI V4.6 and V2 on the 5.93-million-cell T/NK atlas

A read-only, cell-for-cell application of both models at highest-F1 and high-purity operating points, with per-dataset yield, model overlap, and a conservative NK-likeness audit.

259,154 V4.6 highest-F1 predictions	457,628 V2 highest-F1 predictions	0.09% V4.6 strict likely-NK	40 source datasets
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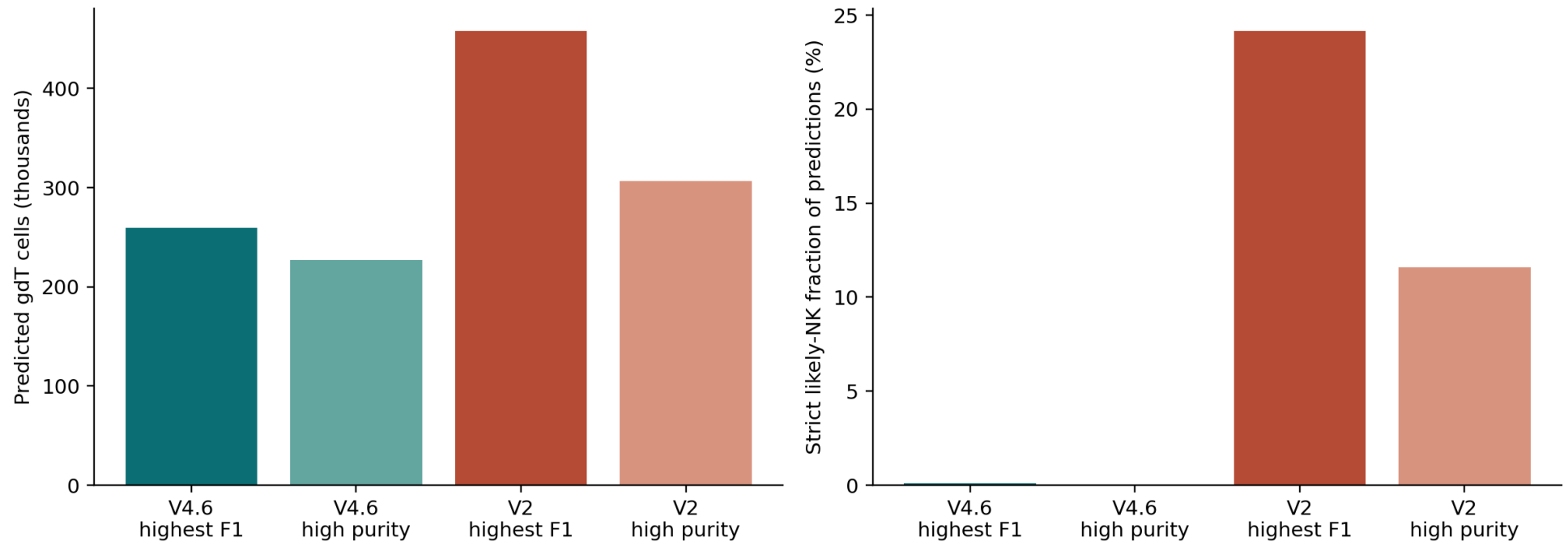
Interpretation boundary. These are deployment counts, not new performance estimates. V4.6 remains development-only. A predicted cell is not a false positive unless independent truth supports that conclusion.

Main result

V4.6 is substantially more selective than V2 on this atlas. At the highest-F1 settings it calls 259,154 cells (4.37%) versus 457,628 (7.71%) for V2. The strict likely-NK audit is 232 cells (0.09%) for V4.6 and 110,489 (24.14%) for V2.

Profile	Predicted gdT	Atlas fraction	Strict likely NK	Strict NK fraction	TRDV+ predicted	TRDC+ TRDV- predicted
V4.6 highest F1	259,154	4.37%	232	0.09%	248,449	5,970
V4.6 high purity	227,189	3.83%	22	0.01%	225,402	1,329
V2 highest F1	457,628	7.71%	110,489	24.14%	244,972	183,187
V2 high purity	306,361	5.16%	35,507	11.59%	230,105	68,696

Whole-atlas prediction yield and conservative NK audit



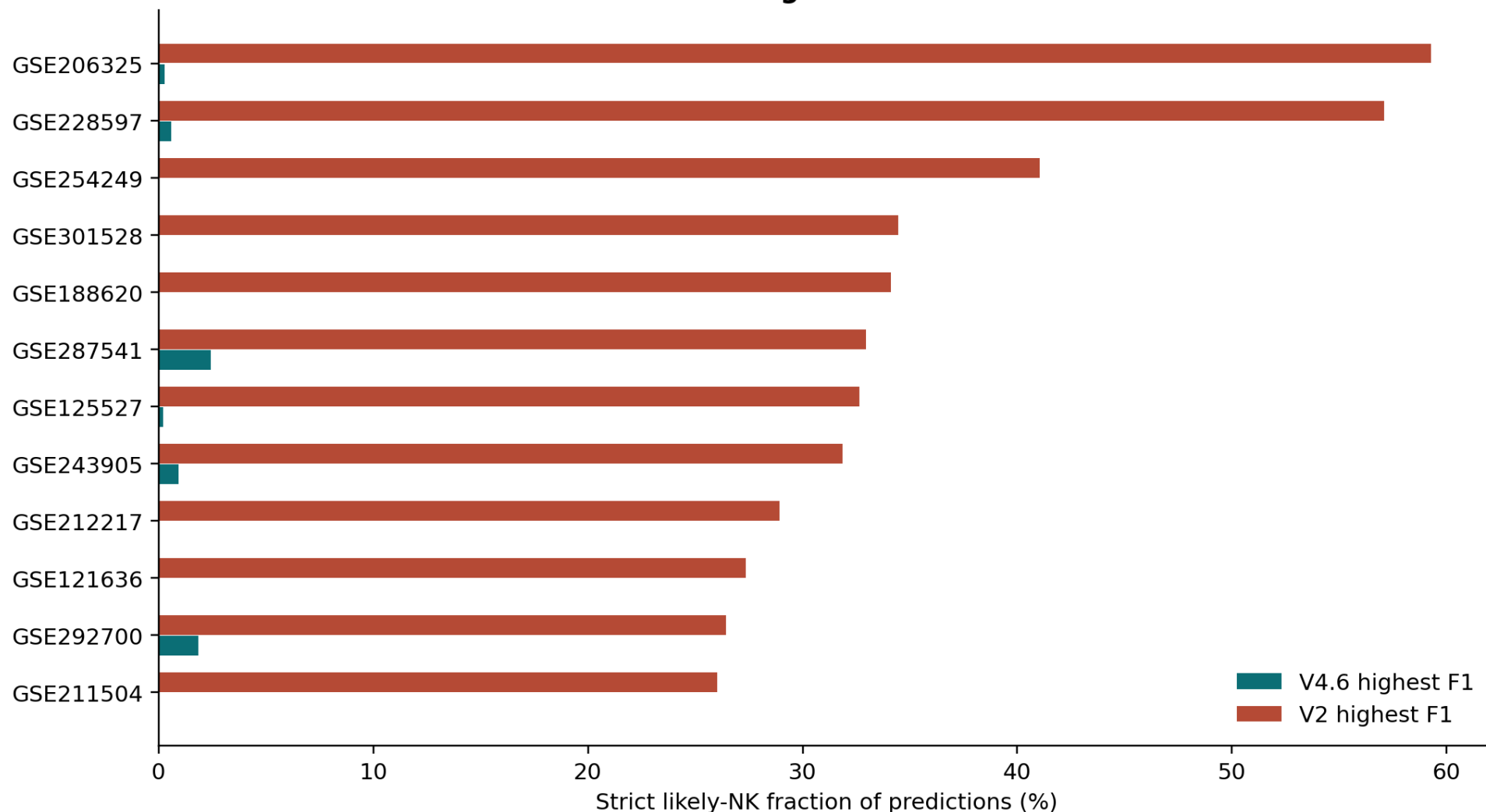
What “likely NK” means

The strict audit requires no TRDV1/2/3 expression and no productive gd-TCR metadata, plus either:

- an explicit source-author NK label; or
- weak CD3 expression (at most one of CD3D/E/G detected), at least one Tier-A innate gene (NCR1, SIGLEC7, SH2D1B), and at least two Tier-A/B genes in total (adding LAT2, SYK, PLCG2, PILRB, CD300C).

NKG7, GNLY, KLRD1, TYROBP, and FCER1G alone are only “cytotoxic ambiguous,” because true gdT cells can express them strongly. Author-NK predictions with TRDV or productive gd-TCR support are exported separately as annotation/receptor conflicts, not counted in the strict NK total.

Datasets with the strongest conservative NK concern

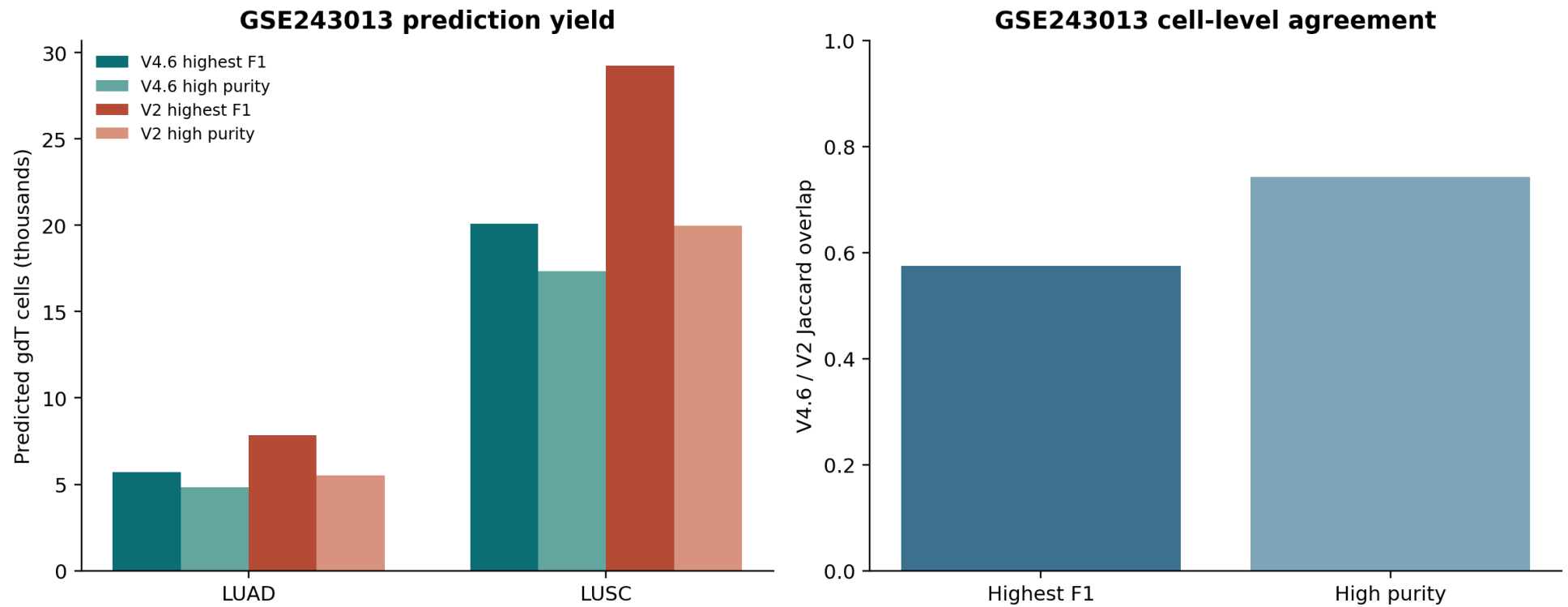


Large LUAD/LUSC project: GSE243013

The corrected metadata contain 759,436 cells: 189,821 LUAD and 569,615 LUSC cells. V4.6 highest-F1 identifies 25,792 cells (3.40%); V2 highest-F1 identifies 37,112 (4.89%). This project has no productive gd-TCR metadata, so expression evidence and model agreement are the available audit signals.

Histology	Profile	Atlas cells	Predicted gdT	Fraction	TRDV+	TRDC+ TRDV-
LUAD	V4.6 highest F1	189,821	5,700	3.00%	5,453	99
LUAD	V4.6 high purity	189,821	4,846	2.55%	4,802	32
LUAD	V2 highest F1	189,821	7,862	4.14%	5,110	2,363
LUAD	V2 high purity	189,821	5,500	2.90%	4,697	718
LUSC	V4.6 highest F1	569,615	20,092	3.53%	19,320	334
LUSC	V4.6 high purity	569,615	17,338	3.04%	17,245	60
LUSC	V2 highest F1	569,615	29,250	5.14%	18,525	8,990
LUSC	V2 high purity	569,615	19,986	3.51%	17,183	2,386

Mode	V4.6	V2	Both	V4.6 only	V2 only	Jaccard
Highest F1	25792	37112	22,942	2,850	14,170	57.41%
High purity	22184	25486	20,302	1,882	5,184	74.18%



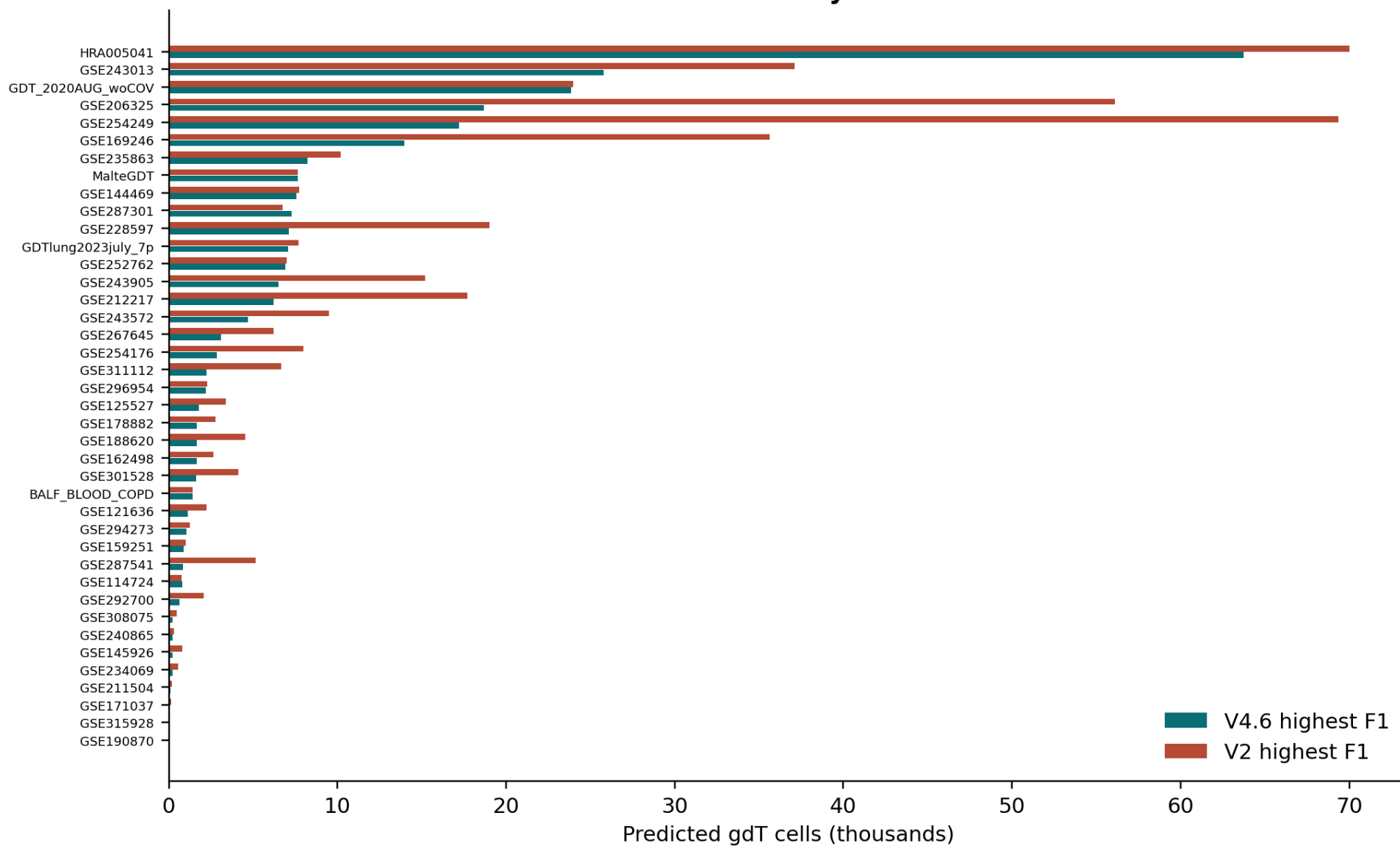
Every dataset

The table below uses highest-F1 for the principal cross-dataset comparison. High-purity values and all audit fields are retained in the canonical CSV tables.

Dataset	Cells	V4.6 predicted	V4.6 strict NK	V4.6 strict NK %	V2 predicted	V2 strict NK	V2 strict NK %
HRA005041	766,639	63742	1	0.00%	70020	598	0.85%
GSE243013	759,436	25792	1	0.00%	37112	4,149	11.18%
GSE206325	501,012	18697	51	0.27%	56114	33,269	59.29%
GSE254249	406,494	17227	0	0.00%	69371	28,482	41.06%
GSE169246	354,373	13987	0	0.00%	35623	8,643	24.26%
GSE243905	350,438	6489	60	0.92%	15197	4,839	31.84%
GSE287301	340,227	7271	1	0.01%	6753	6	0.09%
GSE235863	241,766	8218	1	0.01%	10207	888	8.70%

Dataset	Cells	V4.6 predicted	V4.6 strict NK	V4.6 strict NK %	V2 predicted	V2 strict NK	V2 strict NK %
GSE212217	213,706	6222	0	0.00%	17706	5,120	28.92%
GSE243572	188,217	4693	0	0.00%	9507	988	10.39%
GSE228597	149,812	7119	42	0.59%	19019	10,856	57.08%
GSE188620	123,218	1679	0	0.00%	4516	1,541	34.12%
GSE308075	112,337	232	0	0.00%	459	15	3.27%
GSE287541	110,874	825	20	2.42%	5140	1,693	32.94%
GSE144469	107,068	7559	36	0.48%	7723	688	8.91%
GSE254176	96,953	2832	0	0.00%	7961	1,701	21.37%
GSE267645	92,501	3091	2	0.06%	6218	1,382	22.23%
GSE311112	90,599	2227	0	0.00%	6673	930	13.94%
GSE296954	86,608	2175	0	0.00%	2283	4	0.18%
GSE292700	78,254	654	12	1.83%	2088	552	26.44%
GSE162498	75,880	1641	0	0.00%	2641	8	0.30%
GSE159251	68,898	874	0	0.00%	1019	0	0.00%
GSE178882	67,755	1679	0	0.00%	2780	489	17.59%
GSE315928	66,813	56	0	0.00%	49	0	0.00%
GSE294273	55,744	1054	0	0.00%	1246	85	6.82%
GSE125527	46,396	1792	4	0.22%	3372	1,101	32.65%
BALF_BLOOD_COPD	46,273	1395	0	0.00%	1430	46	3.22%
GSE252762	45,132	6901	0	0.00%	6990	22	0.31%
GSE301528	44,105	1630	0	0.00%	4123	1,421	34.47%
GSE211504	37,244	84	0	0.00%	169	44	26.04%
GSE171037	31,937	60	0	0.00%	128	21	16.41%
GSE114724	28,652	787	0	0.00%	742	0	0.00%
GDT_2020AUG_woCOV	25,904	23877	0	0.00%	23981	22	0.09%
GSE240865	21,680	218	0	0.00%	296	41	13.85%
GSE145926	21,039	218	0	0.00%	802	184	22.94%
GSE234069	20,807	217	0	0.00%	566	48	8.48%
GSE121636	18,793	1129	0	0.00%	2220	607	27.34%
GSE190870	16,753	55	0	0.00%	28	1	3.57%
GDTlung2023july_7p	15,175	7089	1	0.01%	7710	5	0.06%
MalteGDT	7,800	7667	0	0.00%	7646	0	0.00%

Predicted cells in every source dataset



Reproducibility and limitations

- Input: validated corrected atlas, 5,933,312 cells x 27,413 genes.
- Normalization: raw counts to 10,000 counts per cell, followed by log1p, computed in 20,000-cell chunks.
- Frozen thresholds: V4.6 highest-F1 0.894354 and high-purity 0.971830; V2 uses its frozen highest-F1 and annotation-specific high-purity contract.
- No model, calibrator, feature set, or threshold was fitted or tuned on atlas predictions.
- The input H5AD size and nanosecond modification time were unchanged; predictions are stored separately in Parquet.
- Source annotations vary in quality. The strict NK audit is conservative but is still an audit heuristic, not orthogonal cell identity truth.

Canonical outputs: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_6_whole_atlas/`